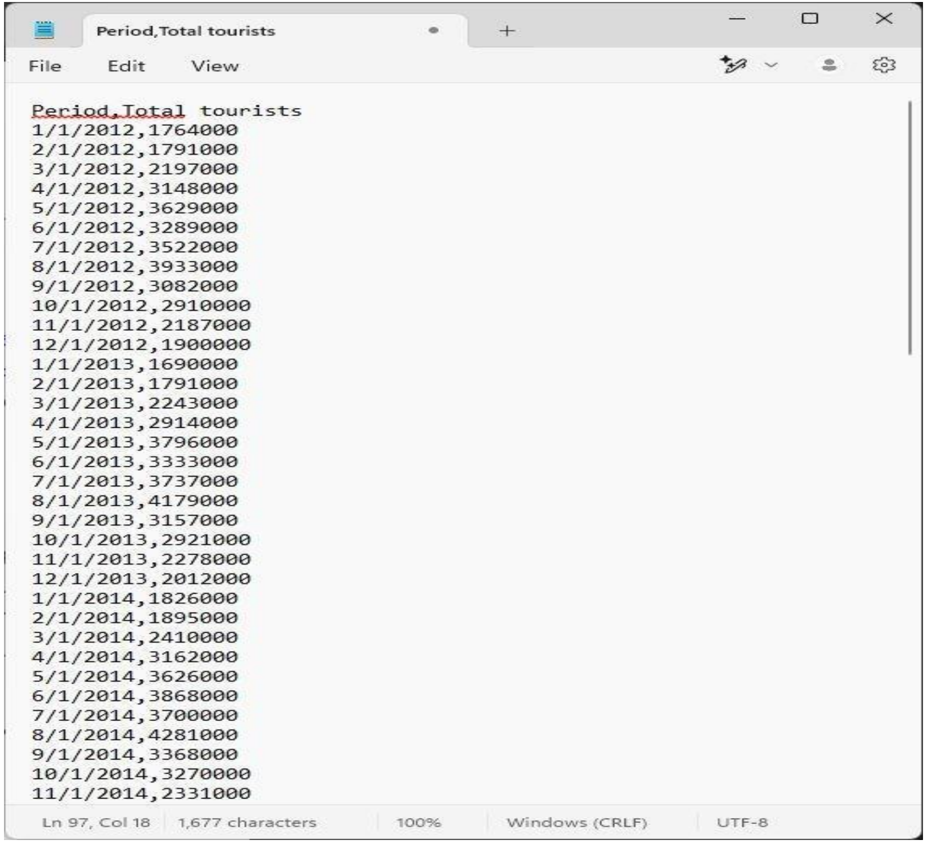
Forecasting Time-Series Data in Power BI

Neil Bryant O. Galindez
Instructor:Joseph A. Vistal
IS 107 Business Intelligence

Step 1: Download the Sample Dataset

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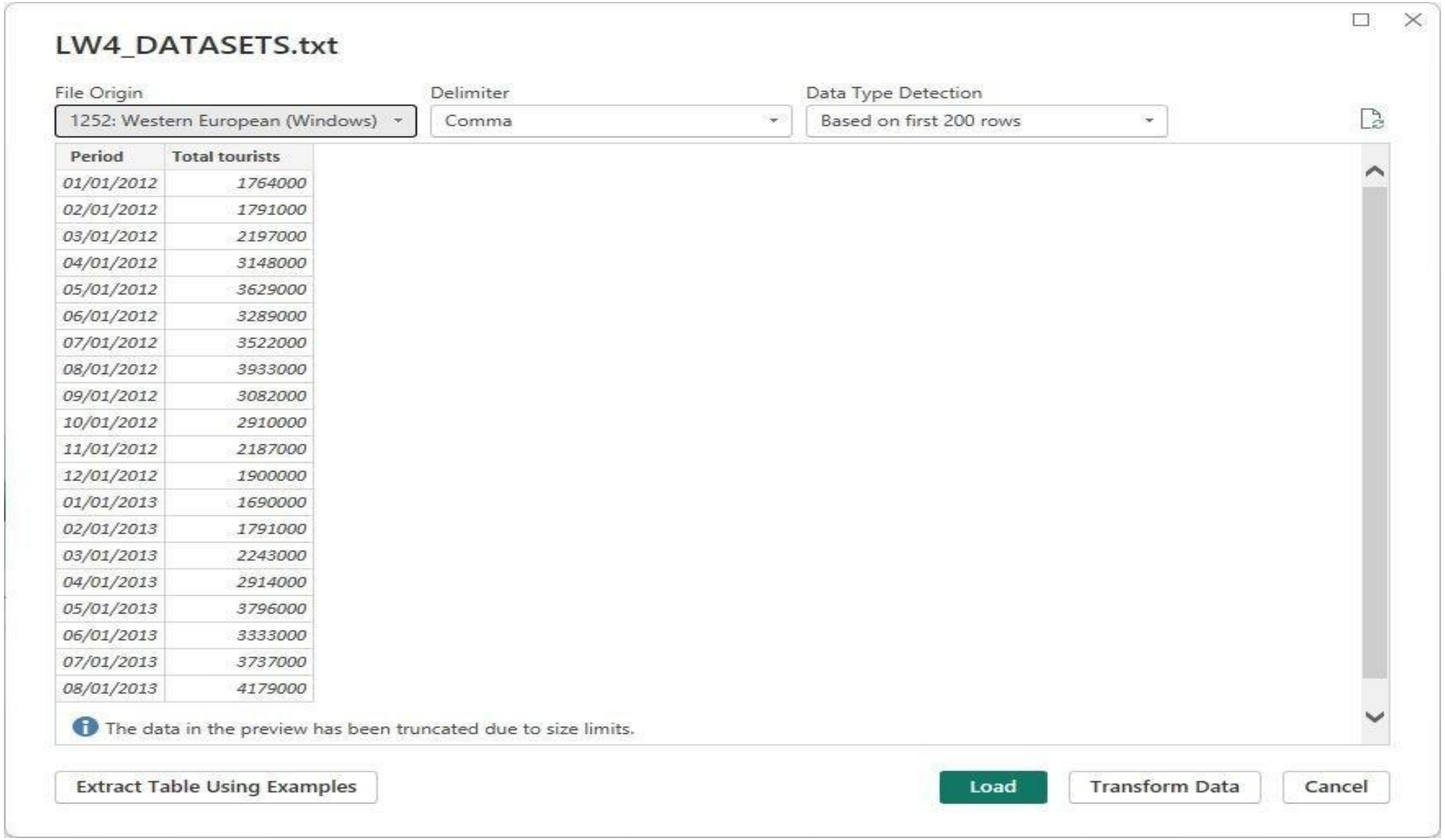
Period,Total tourists
1/1/2012,1764000
2/1/2012,1791000
3/1/2012,2197000
4/1/2012,3148000
5/1/2012,3629000
6/1/2012,3289000
7/1/2012,3522000
8/1/2012,3933000
9/1/2012,3082000
10/1/2012,2910000
11/1/2012,2187000
12/1/2012,1900000
1/1/2013,1690000
2/1/2013,1791000
3/1/2013,2243000
4/1/2013,2914000
5/1/2013,3796000
6/1/2013,3333000
7/1/2013,3737000
8/1/2013,4179000
9/1/2013,3157000
10/1/2013,2921000
11/1/2013,2278000
12/1/2013,2012000
1/1/2014,1826000
2/1/2014,1895000
3/1/2014,2410000
4/1/2014,3162000
5/1/2014,3626000
6/1/2014,3868000
7/1/2014,3700000
8/1/2014,4281000
9/1/2014,3368000
10/1/2014,3270000
11/1/2014,2331000
```

Step 2: Import Data into Power BI



Period	Total tourists
01/01/2012	1764000
02/01/2012	1791000
03/01/2012	2197000
04/01/2012	3148000
05/01/2012	3629000
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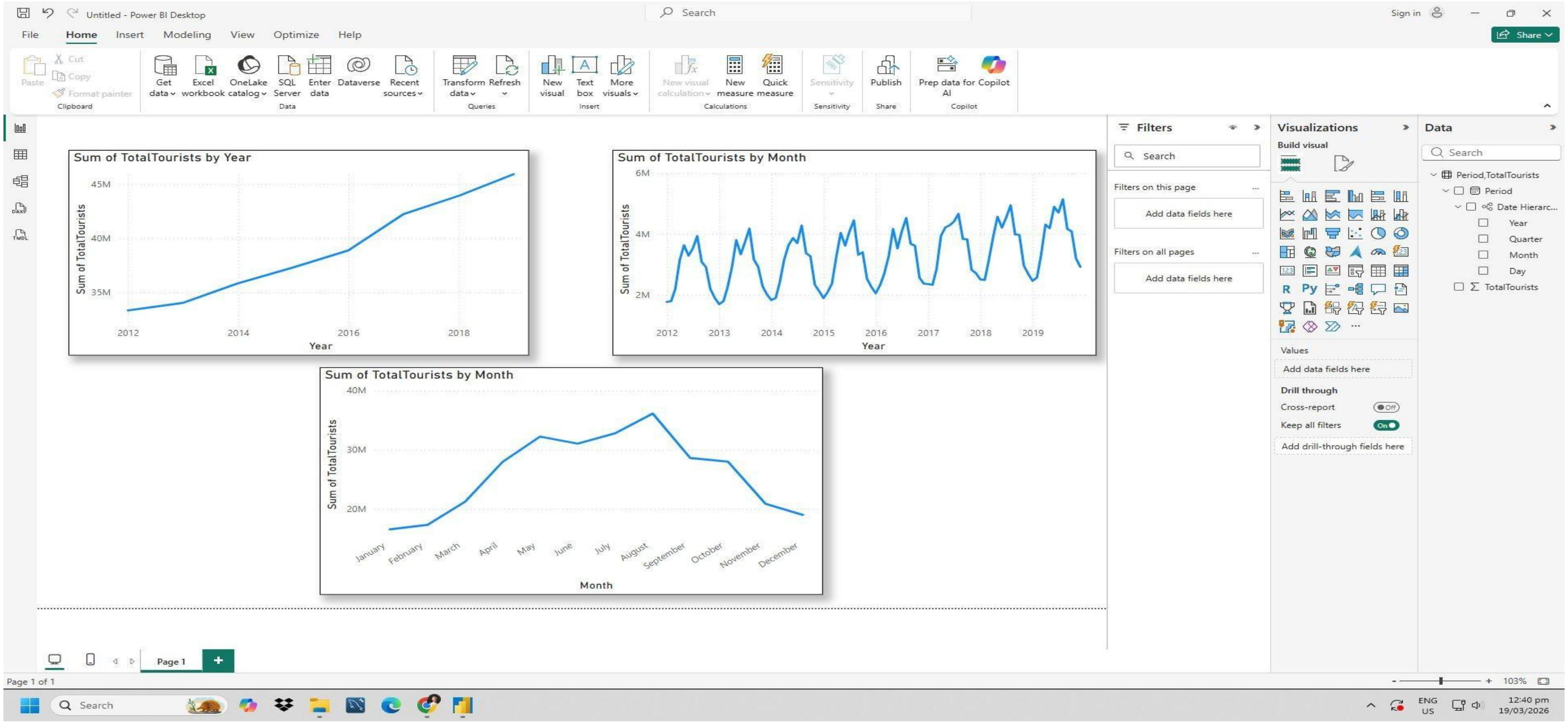
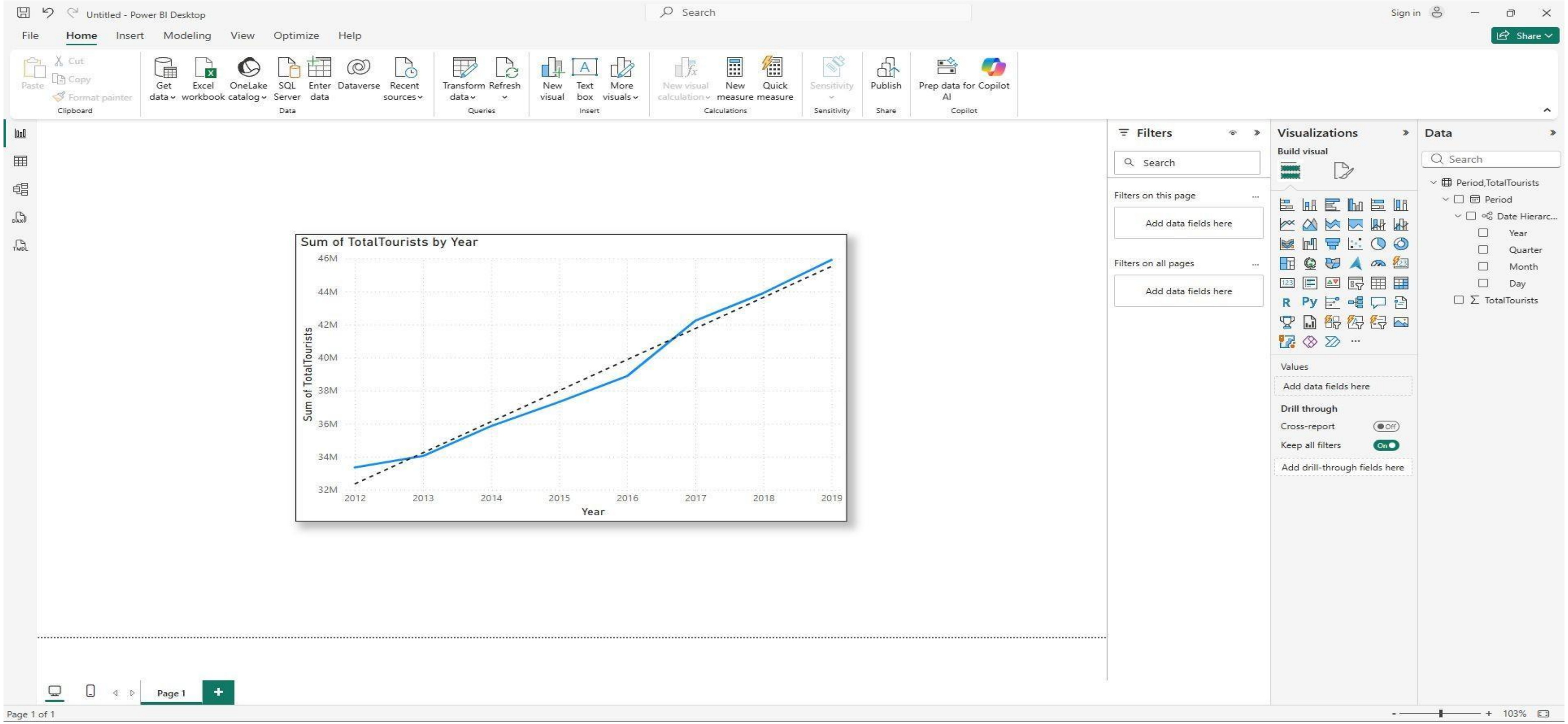
04/01/2013	2914000
05/01/2013	3796000
06/01/2013	3333000
07/01/2013	3737000
08/01/2013	4179000

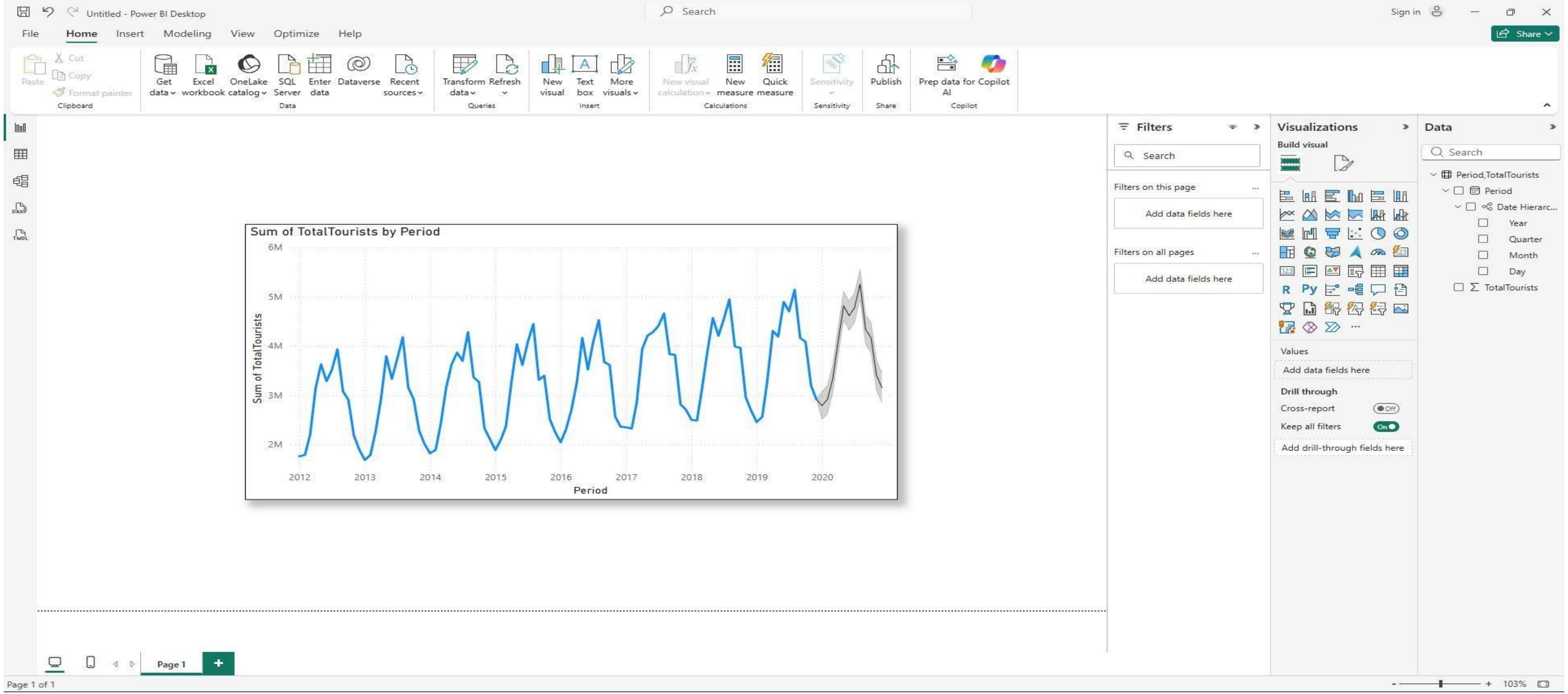
Step 3: Create a Line Chart for Yearly Data



Step 4: Add a Trend Line (Optional but Recommended)



Step 6: Configure the Forecast



Step 7: Analyze the Forecast

1. **Observe the generated forecast for the year 2020.**

The forecast for 2020 shows a continuation of the upward trend and seasonal pattern observed from 2012 to 2019. Based on the historical data , tourist arrivals typically:

- Increase during mid-year (around June–August)
- Decrease toward the end of the year

The forecast assumes that this pattern will continue into 2020, predicting steady growth with seasonal fluctuations.

2. **Note the confidence interval (the shaded gray area). What does its narrowness suggest about the forecast's accuracy for this period?**

The confidence interval (gray shaded area) appears relatively narrow.

This suggests that:

- The model has high confidence in its predictions
- The historical data is consistent and stable
- There are strong repeating seasonal patterns

A narrow confidence interval indicates that the forecast is expected to be fairly accurate, with only small variation from the predicted values.

Step 7: Analyze the Forecast

Compare the forecast to the actual data for 2019.

Looking at 2019:

- Tourist arrivals ranged approximately from 2.4M to 5.1M
- There is a clear seasonal trend
    - Peaks around mid-year (e.g., August ~5.1M)
    - Drops toward year-end The

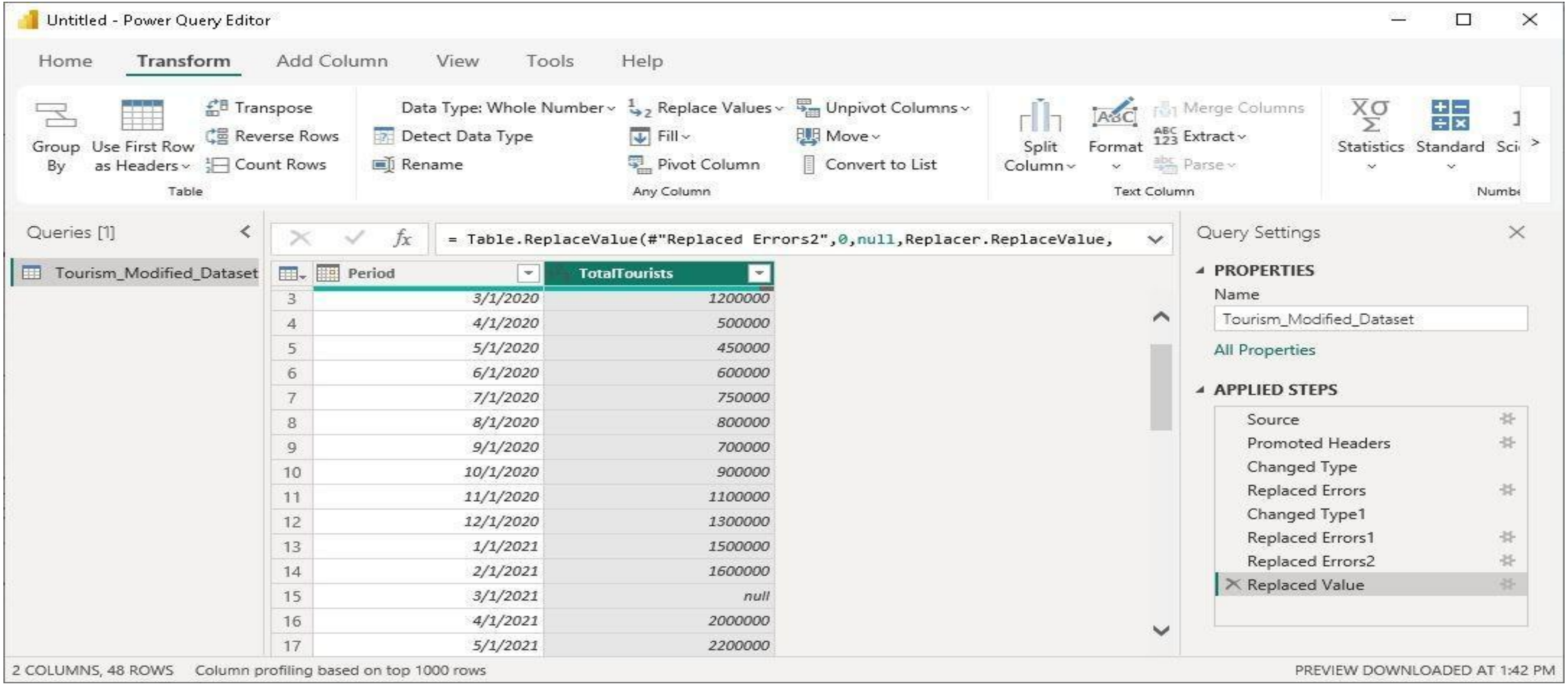
forecast for 2020:

- Follows a similar range and pattern
- Continues the seasonality observed in 2019
- Does not show any sudden drop or disruption

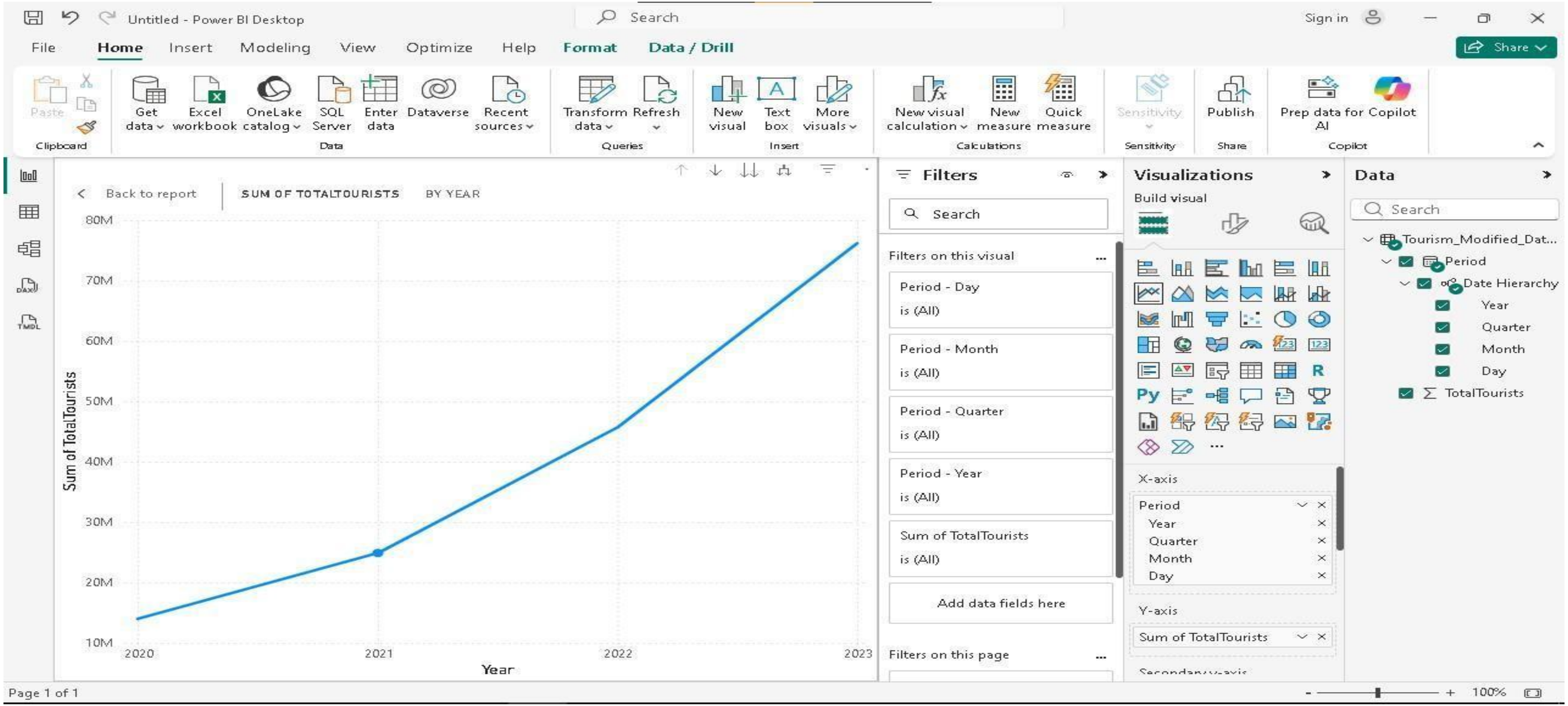
Step 8: Prepare a New Dataset (or use a modified version)

Click here to open new dataset:
https://drive.google.com/file/d/1ndu4kfARLYAQMVBiBNPHHZAcZ7Dx-k9-/view?usp=sharing

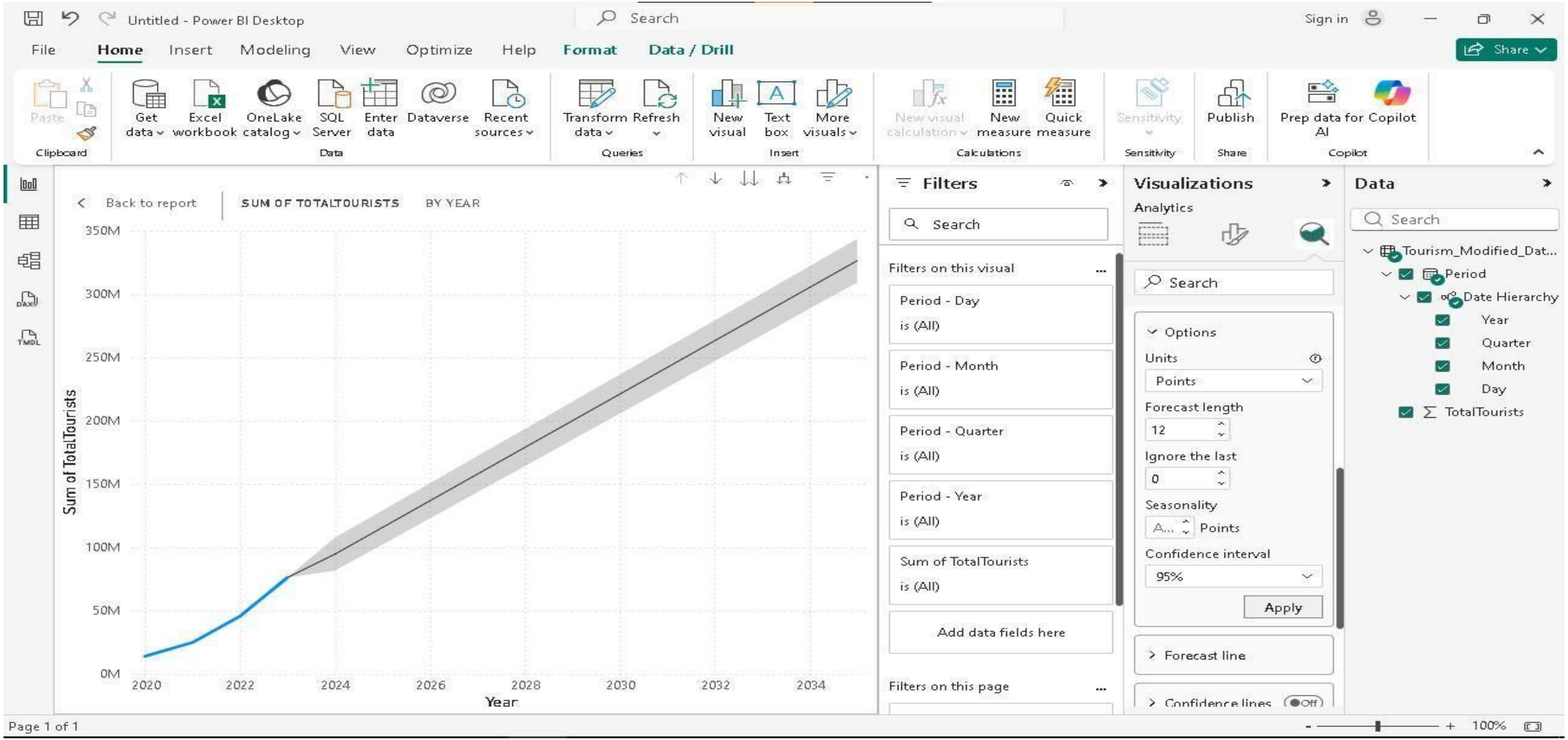
Step 9: Import and Prepare Your New Dataset



Step 10: Create a Line Chart for Your Data



Step 11: Add and Configure the Forecast



Step 12: Analyze and Report Your Findings

1. Describe your dataset: Briefly explain what data you used, its source, and the time period covered.

The dataset used is a tourism dataset containing the number of monthly tourist arrivals. It includes:

- Time variable: Period (date)
- Numerical variable: TotalTourists

The original dataset covers 2012 to 2019, and it was extended with additional data from 2020 to 2023 to simulate real-world conditions. The modified dataset includes:

- Missing values (null)
- Invalid entries (unknown)
- An extreme outlier (10,000,000)
- A sharp drop in 2020 representing a disruption (e.g., pandemic)

Step 12: Analyze and Report Your Findings

This dataset was used to analyze how forecasting handles imperfect and irregular data.

2. Visualize your forecast: Include a screenshot of your Power BI line chart with the forecast applied.

A line chart was created in Power BI using:

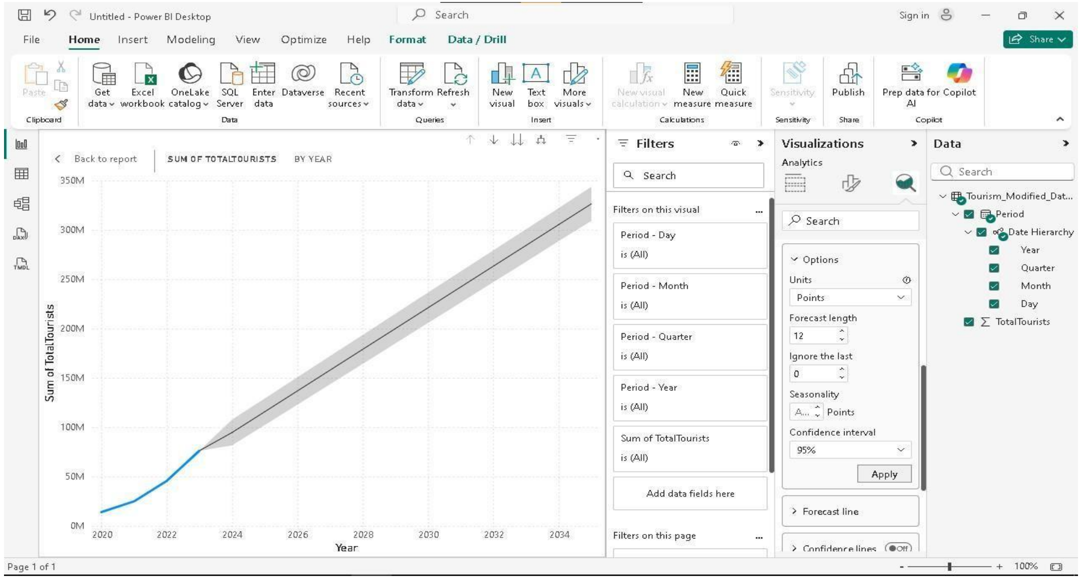
- Axis: Period (Year)
- Values: Sum of TotalTourists

The forecast feature was applied with:

- Forecast length: 12 periods
- Confidence interval: 95%
- Seasonality: Auto

The chart shows:

- Historical data (blue line)
- Forecasted values (black line)



Step 12: Analyze and Report Your Findings

- Confidence interval (gray shaded area)

3. Interpret the forecast:

○ **Discuss the observed trend and seasonality in your data.**

Trend and Seasonality

The data shows a strong upward trend, especially after 2020, indicating recovery in tourism. While the original dataset had seasonal patterns (monthly peaks and drops), the visualization aggregated by year makes seasonality less visible.

○ **Explain the forecast generated by Power BI.**

Forecast Explanation

Power BI generated a forecast that continues the increasing trend observed in recent years. The forecast assumes that the growth pattern will remain stable and extends it linearly into the future.

○

Step 12: Analyze and Report Your Findings

3. Interpret the forecast:

Comment on the confidence interval. Is it narrow or wide? What does this imply about the forecast's reliability for your specific data?

Confidence Interval

The confidence interval is moderately wide and increases over time.

This implies:

- The forecast becomes less certain as it moves further into the future
- There is higher uncertainty due to irregularities in the dataset (e.g., outliers and disruptions)

Step 12: Analyze and Report Your Findings

Interpret the forecast:

○ **Discuss how changing the Ignore last or Confidence interval settings affected the forecast visualization and your interpretation.**

Effect of Changing Settings Ignore

Last:

- When set to 0, the forecast includes recent fluctuations (like the recovery trend)
- Increasing this value (e.g., 3 or 6) ignores recent data, resulting in a smoother but less realistic forecast

Confidence Interval:

- Lower values (90%) produce a narrower range (more confident but riskier)
- Higher values (99%) produce a wider range (less precise but safer)

Step 12: Analyze and Report Your Findings

Interpret the forecast:

○ **How did the forecast handle it? Did it predict the disruption, or did the confidence interval widen significantly?**

Handling of Disruptions

The dataset includes a major disruption in 2020 (sharp drop in tourists).

Observation:

- The forecast did NOT predict the disruption ●

Instead, it:

  - Followed the general upward trend
  - Smoothed out the irregularities
  - Showed increasing uncertainty through a wider confidence interval

This indicates that Power BI forecasting is trend-based and cannot anticipate sudden events.

4. Conclusion: Based on your analysis, how useful is Power BI's forecasting feature for your chosen dataset? What are its strengths and limitations in this context?

Step 12: Analyze and Report Your Findings

Power BI’s forecasting feature is effective for identifying overall trends and future projections when the data is relatively stable.

Strengths

- Easy to use and interpret
- Captures general trends accurately
- Provides confidence intervals to show uncertainty

Limitations

- Cannot predict sudden disruptions (e.g., pandemic effects)
- Sensitive to outliers and imperfect data
- Forecast accuracy decreases over longer periods